RITTICK BARUA, PhD

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SUMMARY

AI strategy and product leader with a hands-on engineering background spanning enterprise transformation, agentic AI architecture and secure production systems. Head of Product & Delivery for an AI platform serving a 1,000+ person European consultancy: lead C-suite decision-making on AI investment, operating models and 6–24-month direction while retaining ownership of product and technical delivery. Former Head of Applied Science; authored the UK Government (DSIT) report *Cyber Security Risks to AI*, with a PhD from the University of Cambridge.

PROFESSIONAL EXPERIENCE

Head of Product & Delivery – **Rud Pedersen Group**, London, UK *Jan 2025 – Present*

Engineering and product delivery* | *Jan 2025 – Mar 2026

- Led the enterprise Chat:R platform, incorporating open-source and commercial models, from proof of concept to production across 20 offices; owned technical requirements, architecture, engineering priorities and release planning; delivered tools for Rud Pedersen's internal operations and client-facing work
- Designed secure agentic workflows integrating Microsoft 365, SharePoint and enterprise systems with per-user permissions; directed isolated agent execution, retrieval and search, tool design, prompt-injection controls and human-in-the-loop actions
- Built and led the AI Systems engineering team, working with developers, QA and business testers; established dev, test and production release gates, CI/CD and operational runbooks
- Owned platform security and operations: investigated and remediated a cross-user OAuth token exposure, commissioned security reviews and a 31-finding audit, and directed Azure infrastructure, identity hardening, incident response and tested recovery procedures

Strategy and executive direction* | *Apr 2026 – Present

- Led C-suite decision-making with the COO, Growth Officer and CFO on enterprise AI direction, investment, risk and business value; established six-month priorities, a 2027 roadmap and a one-to-two-year strategic outlook informed by competition, client expectations and regulation; report fortnightly on progress and decisions required
- Own the AI product vision and portfolio across 20 offices, prioritising products and workflows by their potential to improve consultant output while protecting corporate identity, client confidentiality and human review of consequential work
- Define the commercial model through per-user and per-office pricing scenarios, costs by model, task and tool, and adoption and effectiveness KPIs; directed a change cutting web-search prompt tokens by 42% for equivalent answers
- Shape Group AI Policy and AI governance on EU data residency, model choice, least-privilege integrations, client disclosure, auditability, human approval and release controls; lead executive AI adoption and transformation workshops

Head of Applied Science – **Grant Thornton**, London, UK *Jan 2022 – Nov 2024*

Strategy and thought leadership

- Authored the UK Government (DSIT) white paper *Cyber Security Risks to AI*, incorporating insights from 10+ cross-sector experts to identify 12 critical threat vectors across the AI lifecycle and deliver actionable recommendations to secure AI systems
- Authored an AI-impact report for a global financial conglomerate, surveying 100+ roles across 50+ departments to map near-term disruption and develop a skills-based workforce roadmap
- Led the GenAI adoption programme: defined an enterprise roadmap, prioritised 25+ use cases across five divisions, built governance and success KPIs, and launched eight solutions with £3M+ reported savings
- Led organisation-wide enablement through a champions network, demos to 2,000+ staff and mentoring seven junior data scientists

Applied AI delivery

- Automated 90% of audit report generation with LLMs, freeing five FTE-equivalent capacity while maintaining compliance
- Delivered a computer vision proof of concept for PPE detection on construction sites and an XGBoost EV residual-value model informed by live repair and degradation data

Co-Founder and CEO – **Caritas Research Limited (Non-profit)**, London, UK *Feb 2021 – Mar 2024*

- Co-founded a research and strategy consultancy operating in the UK, India and Bangladesh to evaluate the impact of social interventions

Consultant – **Appian Corporation**, London, UK *Feb 2020 – Jan 2022*

- Proposed and led a resource-allocation application unlocking £1.1M annually of consultant capacity for internal development; led Agile enterprise software delivery and integrations serving 6,000+ concurrent users

Research Engineer – **Thames Water Utilities Ltd**, Reading, UK *Jan 2012 – Dec 2013*

- Secured £120K of industrial funding and led a four-person research team through experimental design and international conference presentation
- Engineered multi-stage reactor systems improving sludge digestion and biogas production by 33–52%, reducing grid-energy dependency

APPOINTMENTS

Visiting Researcher – UCL Hospital, Surgical Biotechnology, London, UK *Mar 2025 – Mar 2026*
Visiting Research Fellow – Manchester Metropolitan University, Manchester, UK *Apr 2024 – Apr 2026*

EDUCATION

PhD, Chemical Engineering and Biotechnology – University of Cambridge, Cambridge, UK *Oct 2014 – Sep 2018*

- Fully funded doctoral research; peer-reviewed paper and international conference best-poster award; novel flow model for complex fluids and oil-well integrity

BEng (Hons), Chemical Engineering (First Class) – University College London, London, UK *Oct 2007 – Jun 2011*

SELECTED PUBLICATIONS

- *Cyber Security Risks to Artificial Intelligence*, UK Government Department for Science, Innovation and Technology, 2024
- *Guide to Artificial Intelligence in Jersey's Finance Industry*, Jersey Finance, 2024

SKILLS AND TECHNOLOGY

- **AI and LLM systems** – RAG, Azure AI Search, pgvector, agentic workflows, MCP servers, prompt-injection containment, human-in-the-loop approvals, model governance and residency, evaluation and cost control
- **Security and identity** – Azure and source-code security reviews, AI threat assessment, Entra ID, OIDC/OAuth, Microsoft Graph, per-user token binding, secure CI/CD, CodeQL, secret scanning, GDPR and EU AI Act readiness
- **Machine learning** – U-Net segmentation, CNNs, object detection, XGBoost, SHAP, time-series forecasting and classification
- **Languages and libraries** – Python (NumPy, SciPy, Pandas, scikit-learn, PyTorch, TensorFlow, OpenCV, FastAPI, Playwright), SQL, JavaScript/TypeScript (Node, Express, React), Bash, MATLAB
- **Cloud and platforms** – Azure OpenAI, AI Search, Document Intelligence, Functions, App Service, Key Vault, Monitor, Docker, GitHub Actions, Bicep, MongoDB, Redis, n8n, Claude Code and Codex